BUILT BY YOUNG PEOPLE

FOR YOUNG PEOPLE

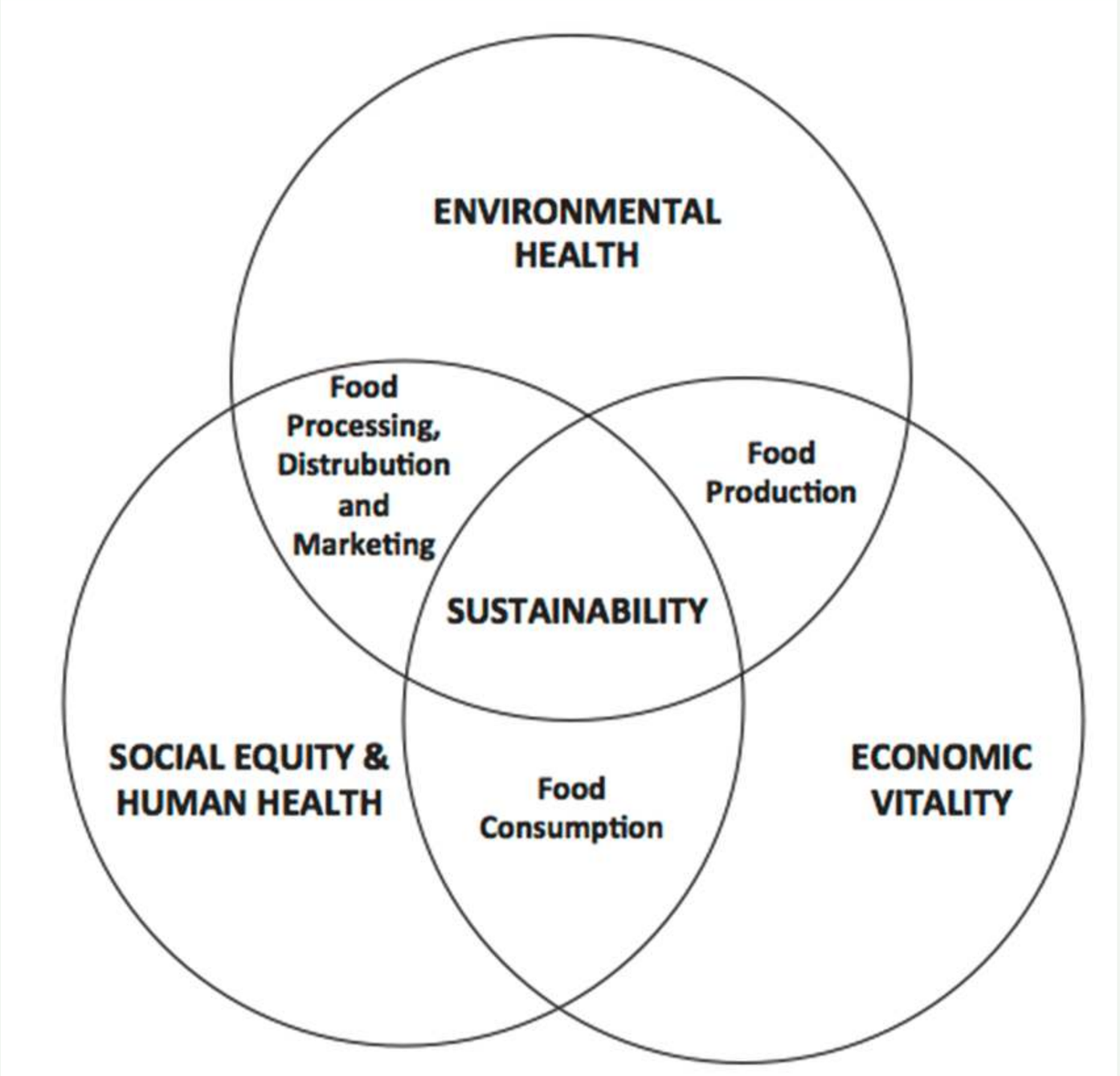
OY4CCurriculum

Topic 10: Food Management

Students answer:

What do you know about food management?





Allievi, F. (2017/12/14) MEAT CONSUMPTION AS A WICKED PROBLEM: EVIDENCE FROM DATA AND POLICIES.

Aims

- Foster an in-depth understanding of food management by exploring its connections with the climate crisis.
- Discover solutions to poor food management and how to get involved.
- Deliver these crucial food management concepts to students in a way that. resonates with their daily lives, ensuring practical relevance and applicability.
- Encourage students to make a positive impact by using their knowledge of food management to address real-world climate and social challenges.

Key Terms

Sustainable Agriculture

A mode of farming that uses practices that create a net-zero impact regarding the harm done to the environment since, by definition, to be sustainable means to maintain the same.



Rawles, S. (10629) '© WWF-Canon / Simon Rawles', Available at: https://www.worldwildlife.org/industries/sustainable-agriculture).

Behrend, G. (2020) Regenerative vs. sustainable agriculture: What's the difference?, Savory Institute. Available at: https://savory.global/regenerative-agriculture-sustainable-agriculture-differences-holistic-management/ (Accessed: 15 June 2023).



Regenerative Agriculture

"Strives to work in cooperation with the ecosystem and strengthen it, thus restoring and benefiting the environment in the process."

AdobeStock_462557017 2023, Available at: https://ik.imagekit.io/3hgudr3tlr/wp-content/uploads/2023/12/AdobeStock_462557017.jpeg,

Pandolfo, Georgie, Food for Thought...and the World, Published: Feb 2022. https://docs.google.com/document/d/1FzNN_JJjld-lG7W3lehl874hjTIct2wqgOAu7TegDjA/edit?usp=sharing

Land & Soil Degradation

The process by which an area of land is overworked to the point that it can no longer successfully produce crops.



via Earth.com

Pandolfo, Georgie, Food for Thought...and the World, Published: Feb 2022.

Introduction

“Food and a habitable planet are basic needs to survive.
We shouldn't have to choose between one or the other.”

-Ag for Equity



Niles, M.T., Ahuja, R., Esquivel, J., Mango, N., Duncan, M., Heller, M.C., & Tirado, C. (2017) Climate Change and Food Systems: Assessing Impacts and Opportunities.

Introduction

Food and a healthy planet are much more interconnected than you may think.

- Food is a human basic need...
  - But, it also significantly impacts the environment

- Food consumption's environmental impact is crucial in assessing the climate crisis.

Food Waste

The Big Picture of the Problem



Credit: United Nations, 'International Day of Awareness on Food Loss and Waste Reduction' (2023)

- Leftover food has a significant impact on the planet.
- Daily food waste is the majority in landfills.
- EU food waste
  - 58 million tonnes annually, 131 kg/inhabitant, market value 132 billion euros

The Big Picture of the Problem

Executive Summary

The global population exceeded seven billion people during 2011 and is predicted to reach 9.3bn by 2050, with a projected increased food demand of 50-70%. Against this backdrop of rising demand, 868 million people are chronically under-nourished, equating to one in eight people worldwide. At the same time, it is estimated that over one third of all food produced globally for human consumption goes to waste. In the UK, 5.6 million people live in deep poverty, where basic food provision is a daily challenge; yet at the same time, 15 million tonnes of food is wasted annually, with nearly half discarded within UK households. Therefore, reducing the scale of losses and waste throughout the entire food system is a crucial step towards improving global food security. This report provides an independent assessment of the issues around food waste in developing and developed countries and suggests a number of potential future research priorities across the food supply chain.

- Landfilled food rots, releasing methane (CH4).
  - Methane is a greenhouse gas that contributes to global warming.
- Dangerous consequences of throwing away food.
- Worldwide: one-third of produced food is uneaten.
- Food waste is the third-largest greenhouse gas emitter.

Bond, M., Meacham, T., Bhunnoo, R. and Benton, T., 2013.
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Solutions – Reducing Food Waste

- ReFED tackles food waste in the U.S. through established tactics.
- FarmLink focuses on preventing food waste.
- OzHarvest in Sydney, Australia, fights food waste through education, rescue, and sustainability.

the farmlink project

ReFED
Rethink Food Waste
Through Economics and Data

OZHARVEST

Solutions – Reducing Food Waste

- World Vegetable Centre in Tainan City, Taiwan, raises awareness about the role of vegetables.
- Imperfect Foods, a grocery delivery service, reduces food waste by selling "unaesthetic" food.
- Restaurants partner with local food banks to donate unused food to those in need.

IMPERFECT FOODS

50

Sowing seeds, meeting needs

World Vegetable Center

1973-2023

Supporting Questions

- What does this mean for communities receiving their support?
- How can this be implemented on a larger scale? Policies?
- What can I do about it?

Solutions – Reducing Food Waste

On an individual level, there are various ways to reduce food waste at home:

1. Buy only the necessary amount of food to avoid excess.
2. Learn proper food storage techniques to prevent spoilage.
3. Instead of discarding food, consider donating it to those in need.
4. Unusable food items like waste oils can be converted into fuels.
5. Compost excess food for environmental benefits.
6. As a last option, dispose of food waste responsibly.

Systemically, promoting regenerative or sustainable agriculture minimizes food waste by focusing on efficient production methods.

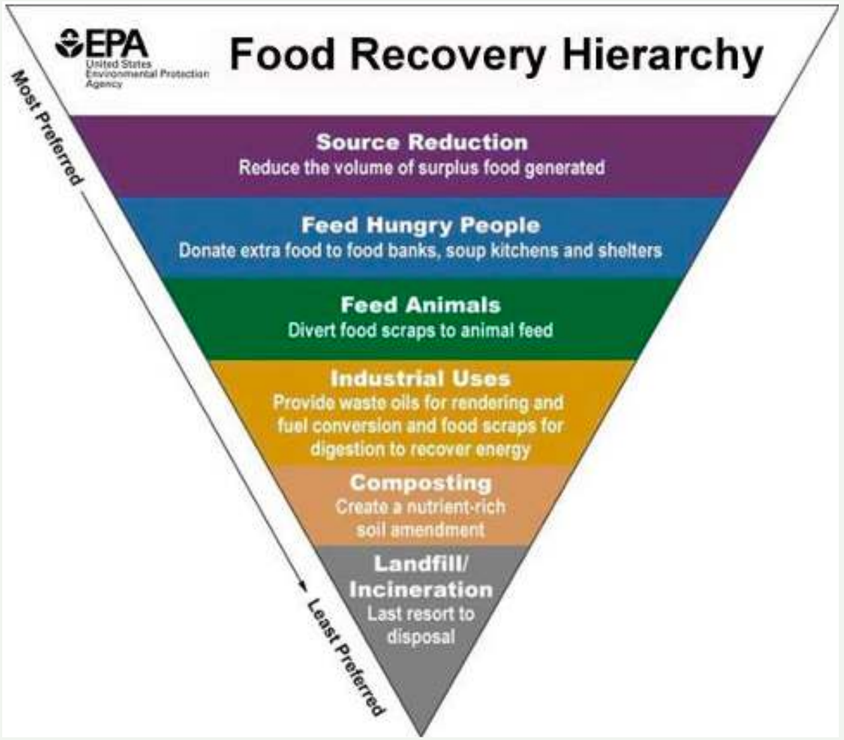


Image source: “Food Waste.” The Nutrition Source, 26 Apr. 2017, www.hsph.harvard.edu/nutritionsource/sustainability/food-waste/. Accessed 31 Jan. 2023.

Regenerative Agriculture

vs.

Sustainable Agriculture

Sustainable Farming

Aims for a "net-zero" environmental impact in food production.

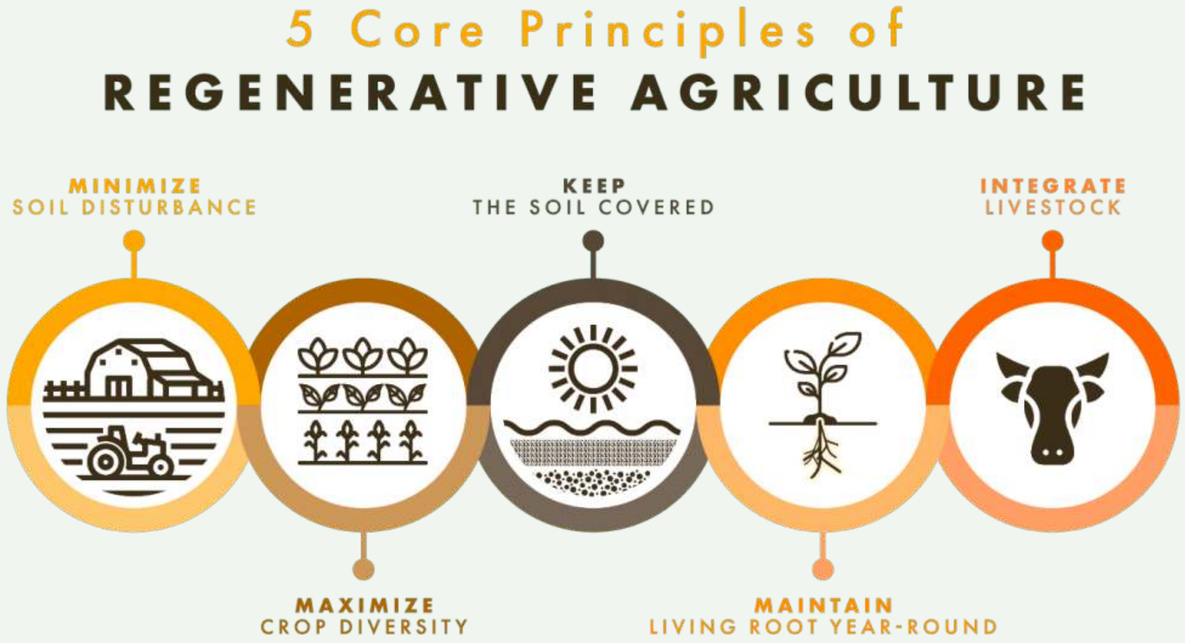


Rawles, S. (10629) '© WWF-Canon / Simon Rawles', Available at: https://www.worldwildlife.org/industries/sustainable-agriculture).

Regenerative Agriculture

Unique approach with multiple benefits

- Fewer greenhouse gas emissions.
- Less invasive farming practices.
- Enhances soil health and strengthens its microbiome.
- Traps in more nutrients.
- Increases biodiversity in the surrounding environment.
- Economically viable for farmers.

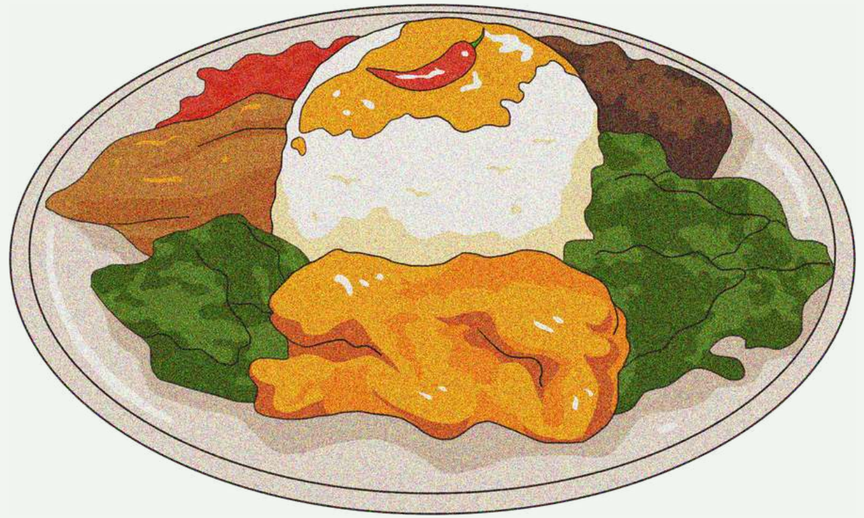


credit: Grainmillers.com

Intersectionality in Food Management

Intersectionality & Resources

- Individual efforts are crucial for countering policy gaps.
  - However, a sustainable diet is not universally accessible.
- Low-income, marginalized, and people of color communities face health and economic disparities.
- Disproportionate vulnerability to unsustainable food practices.
  - Issues include water pollution and climate crisis impacts.
  - Negative effects on health within these communities.



Food Justice Movement



Bobby Seale and Huey P. Newton
Black Panther Party national chairman Bobby Seale (left) and defense minister Huey P. Newton.;AP

- Curran and González (2011), “Communities of color face disproportionate health issues.”

- The Food Justice Movement emerges in response to injustices.
- Black Panther Party influenced its early development.
  - Implementation of food programs targeting low-income people of color.
  - Notable initiative: Free Breakfast for Children Program.
    - The program has an immediate positive impact on black youth.
- Illustrates the importance of intersectional approaches to food management.

Principles for Food System Evaluation



Professor Bhavani Shankar via University of Sheffield, Department of Geography

Professor Bhavani Shankar introduces principles for food systems evaluation.

- Focus on efficiency, meaningful representation, and recognizing rights.
- Emphasis on addressing disparities in food management.
- Calls for transformative action and collective responsibility.

Youth Climate Activist Examples

Farmlink Founders

- Founded by Aidan Reilly, James Kankoff, and Ben Collier during the COVID-19 peak in 2020.
- Response to supply-chain issues and increased demand for food banks.
- Connects local farmers with excess produce to communities in need.
    - Started with a delivery of 10,800 eggs to Westside Food Bank.
    - Rapidly expanded, distributing over 70 million pounds of food.



Image: Debeck, O. (2020). Available at: https://variety.com/2020/biz/news/farmlink-foodbanks-farmers-aidan-reilly-will-collier-1234606160/. (Accessed 7 March 2025).

https://www.foodrecoverynetwork.org/blog/2021/4/13/celebrating-a-year-of-food-recovery-with-the-farmlink-project

Farmlink Founders

- Bridge the gap between surplus farms and food banks.
- Addressing food insecurity reduces carbon emissions.
- Fosters the next generation of changemakers.
- Actively works towards a more sustainable and accessible future.
    - Promotes efficiency in farming and healthier produce access.

- Learn more at farmlinkproject.org



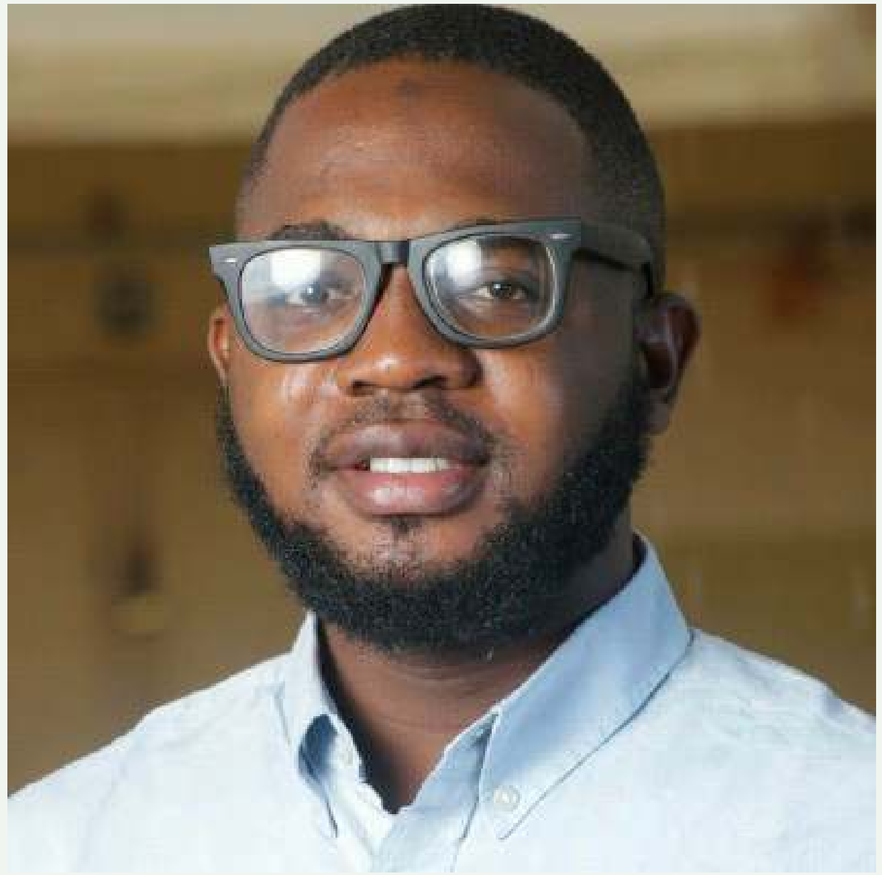
Image: Debeck, O. (2020). Available at: https://variety.com/2020/biz/news/farmlink-foodbanks-farmers-aidan-reilly-will-collier-1234606160/. (Accessed 7 March 2025).

https://www.foodrecoverynetwork.org/blog/2021/4/13/celebrating-a-year-of-food-recovery-with-the-farmlink-project

Azeez Salawu

- Youth Leader for the UN Food Systems Summit Dialogue.
- Member of Youth Liaisons supporting the UN Food Systems Summit.
- Member of the Nigerian Civil Society Framework on Paris Agreement and SDGs.
- Engages at governmental, policy, community, and citizen levels.

“I pledge to use my position to advocate against zero hunger and malnutrition.”



Credit: https://www.gainhealth.org/azeez-salawu

https://www.foodrecoverynetwork.org/blog/2021/4/13/celebrating-a-year-of-food-recovery-with-the-farmlink-project

Conclusion

Conclusion

- Managing food involves a complex process, from growth to disposal.
  - Every step contributes to energy production, waste, and environmental impact

- Regenerative farming
  - A holistic approach to farming that gives back to the environment.
  - It goes beyond just eliminating harm and focuses on actively benefiting the environment.
  - Involves various techniques like intercropping, crop rotation, no-till farming, contour farming, agroforestry, composting, and more.
- Emphasizes the importance of considering each step to avoid environmental damage in the food management process.



Thank you for learning about Food Management!

Skills Learned

- Understanding the relevance of agriculture and food management in the broader picture of the climate crisis.
- Recognizing areas for potential improvement in our current food systems.
- Having the background knowledge concerning food systems to make eco-conscious decisions regarding food, its distribution, and use.



Job Ideas in the Field

- **Research Scientist:**
  - Conducts experiments, analyzes data, and contributes to scientific advancements in a specialized field through research and innovative discoveries.

- **Agricultural Policy Analyst:**
  - Evaluates agricultural policies, conducts research, and offers recommendations to enhance farming practices and support sustainable agricultural policies.



Discussion Time

Action

Design and/or build a regenerative urban garden based on what you learned



Goal: Allow the students an opportunity to apply the knowledge they've learned to create their own community garden. This can be a design for a hypothetical garden, or if applicable and resources allow, this can be the blueprint for a real garden that the students will maintain together.

Instructions

- Determine the physical dimensions of your garden (and a potential location if you will actually be building this garden).
- Reviewing the various regenerative farming methods detailed in this curriculum, decide which you would like to use and how you will put them into practice.
- Conduct online research as necessary (e.g. which produce is best to grow in your local area, planting and harvesting seasons, if the plants grow well together, etc.).



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